embedded_vector_databases_edge (Initial Draft)

Domain: Edge Iot | Document ID: embedded_vector_databases_edge_v1
Specification Standard: IEEE Capstone Evaluation Corpus

Executive Overview:

Embedded vector storage using Qdrant and SQLite-VSS on memory-constrained single-board edge hardware.

Methodological Verification & System Lineage:

This specification is catalogued in IntentCloud for intent-aware hybrid retrieval.
Text chunks are indexed into 384-dimensional dense semantic vectors and 1,000,003-dimensional sparse feature-hashed token projections.
Candidate pools are unified using scale-invariant Reciprocal Rank Fusion (k=60).
Explainable citation attribution is extracted using cross-encoder cross-attention.